# CodeSport

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**Welcome to another instalment of CodeSport. This is a special Diwali Edition in which we feature 20 interesting programming questions often asked in interviews.**

As you know, every month we discuss one specific topic in programming and feature a couple of coding questions related to it. However, we will take a break this month since many readers have asked me for programming questions that they might find useful to practice and warm up for prior to an interview. So here are 20 interesting programming questions covering various topics in algorithms and data structures. I will leave these questions open for our readers to answer, and we will feature their solutions in next month's column. Let's get started with the easier ones.

(1) Can you have both const and volatile qualifiers applied to a single declaration in C? If so, can you give an example? If not, why not?

(2) Consider the following code snippet:

```
main()
{
    int a, b;
    a = foo();
    b = foo ();
    printf("a = %d  b =%d \n", a, b);
}
```

You do not know anything about function 'foo' except that it returns an integer.

Are the values of *a* and *b* printed by the *printf* function the same? If not, can you give an example of function 'foo' which will cause *a* and *b* to be different?

(3) Given a word, can you find all its anagrams? If you were asked to do this for only one word, what would be your solution? If you were asked to find the anagrams for 10,000 words, would your solution change?

(4) What is the time complexity of searching for an element in a...

a) Linked list containing N elements
b) Hash table containing N elements
c) Binary search tree containing N elements
d) A binary heap containing N elements
e) A d-heap containing N elements

(5) Consider the following code snippet:

```
int find_fib (int N)
{
    assert (N > 0);
    int f1 = 1;
    int f2 = 1;

    if (N < 2)
        return 1;

    return (find_fib(N-1) + find_fib((N-2));
}
```

Given that *find_fib* is called from *main* with N as 25, how many total calls are made to *find_fib*?

## Sorting/searching/string manipulation problems

(6) You are given an array of 2n+1 integers. You are told that except for one element, all other elements have a duplicate in the array. Can you find the one element that has no duplicate in the array? What is the time complexity of your solution?

(7) You are given an array A of N integers. A majority element M is an element that appears more than N/2 times in the array. For instance, given the array of integers, 10, 6, 10, 3, 10, 10 is the majority element. Given the array 10, 6, 10, 3, 10, 21, 5, there is no majority element. For the given array of N integers, write an algorithm to find whether a majority element exists and if so, what is it? The algorithm should have a worst-case complexity of O (N).

(8) You are given two strings, S1 and S2. The maximum length of either of the strings is N characters. You need to create a new string containing only those characters that appear in both the strings. Remember that the trivial solution is O (N^2) wherein we compare each character of the first string to every character in the second string. Can you come up with a better solution?

(9) You are given a sorted sequence of distinct integers, a1, a2, a3 .... aN. Give an algorithm to determine whether there exists an index *i* such that the element at the $i^{th}$ position is equal in